

Emotional Support Companions in Virtual Reality

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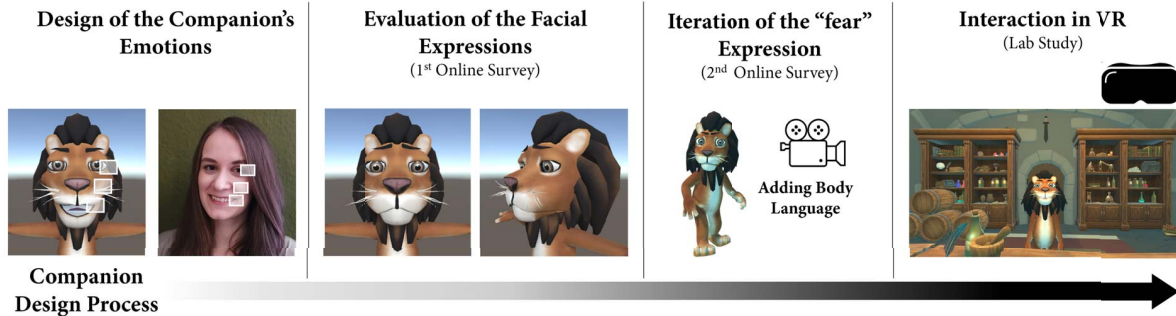


Figure 1: This figure demonstrates how we designed and evaluate the companion's emotions. First, we followed the instructions of the Facial Action Coding System (FACS) [4]. For example, a happy face is characterized by narrowed eyes, raised cheeks, pulled-back lips, and expressed teeth. After that, we evaluated the recognizability of the facial expressions and also tested the interaction with the companion in VR to check its impact of the users' emotions.

ABSTRACT

According to social psychological models, the presence of another person or even a virtual character in stressful situations can have stress-reducing effects. Thereby, the outcome can depend on the congruency between one's mood and the perceived mood of the other person. This dependence guides the design of VR applications for stress reduction, which use different virtual companion designs depending on the emotional state of individual users. This paper describes an ongoing design and development process towards such an emotionally supportive companion and shares initial results concerning the perception and stress-reducing effects of positively- and negatively-minded companions.

Index Terms: Human-centered computing—Visualization—Empirical studies in HCI

1 INTRODUCTION

Social support of others can help reduce stress and improve well-being [3]. As other people are not always available, virtual characters or companions could help instead [8]. A 'companion' is a virtual character who accompanies players in virtual environments. It is an agent or non-player character (NPC), so therefore controlled by the computer [16]. Companions can support the players' behavior and feelings and show what certain feelings are appropriate in the specific situation [12]. Hence, companions make the players adapt their emotions and behavior [5], or evoke empathy by showing emotional facial expressions [15]. Liszio et al. used penguin companions in VR to make children not feel lonely while preparing for an MRI [10]. Felnhofer et al. show a stress reduction through the presence of a virtual character, in a subsequent stress situation [6]. But in what mood must the Companion itself be, to provide optimal emotional support? According to *Emotional Similarity*, supporters

are expected to be in a similar mood as the one obtaining support to give them a feeling of compassion [14]. In contrast, according to the *Emotional Contagion* theory, supporters might be in a distinct mood to infect the one with their positivity and therefore cheer one up [9]. Hence, we formulated the following research question: *What effect does the interaction with a positively- or negatively-minded virtual companion have on the users' emotion?*

2 DESIGN OF THE COMPANION'S EMOTIONS

We designed two companions: (1) A happy companion that displays positive emotions, and (2) a stressed companion that displays negative emotions. First, we decided not to choose a human companion, to avoid the uncanny valley effect [11]. We, therefore, picked an animal companion, as studies could show similar social effects of non-human characters [15]. Hence, we chose a 3D-model of a humanoid lion¹, as the character should still be human-like to enable *Emotional Contagion* [13]. We used animations from Mixamo and modeled the facial expressions after the Facial Action Coding System's (FACS) emotions happiness, fear, and sadness [4] (Figure 1). We also gave the companions voices, by writing a positive and negative script, which contain validated sentences for the targeted elicitation of the respective emotions [1].

2.1 Evaluation of the Facial Expressions

To make sure that the different companions were perceived correctly, we conducted an online survey in which we checked the lion's facial expressions. We presented five different facial expressions: fearful, sad, neutral, two versions of happy varying degrees of severity. First, the participants were shown two face pictures, one frontal and one from the side (Figure 1). Then, they were asked to choose which emotion was expressed from a prescribed selection, based on the universal emotions by Ekman [4], extended by the options *neutral* and *other*. Furthermore, the participants were asked to rate how sure they felt about their decision. 44 participants (19 female; age: 20 - 57; $M = 30.1$, $SD = 10.2$) filled out the online

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¹<https://assetstore.unity.com/packages/3d/characters/animals/toon-humanoid-lion-pbr-129773>

Table 1: Means of the DEQ for the positive (PC) and negative companion (NC). Significant p -values marked with a * ($p < .05$).

Emotions (DEQ)	PC		NC	
	M		p	
Fear	5.28	13.5	< .001*	
Anxiety	6.09	20.2	< .001*	
Sadness	5.24	9.90	< .001*	
Relaxation	23.1	11.2	< .001*	
Happiness	21.0	14.1	< .001*	

survey. Results showed that the neutral (75%), less expressive happy (55%), expressive happy (80%), and sad (91%) facial expressions were recognized sufficiently well. Fear was mistaken as surprise expression (39%) as often as it was identified correctly (39%). Based on these findings the expressive happy expression was used for further development and improvements were carried out on the fear expression.

2.2 Iteration of the "fear" expression

Additionally, we evaluated the improved fearful expression. Since studies indicate that the recognition of especially fearful expressions improves when presented combined with the character's respective body movement [2], we used a short video of the companion including the corresponding body language. The procedure was equal to the first survey. 25 people (10 female; age: 22 - 55; $M = 30.9$, $SD = 11.0$) participated. 12 (48%) participants identified the companions emotion as fear or insecurity while two (8%) participants recognized surprise. Yet, seven (28%) participants did not recognize any emotion (neutral). Subsequently, the behaviour of the companion was further adjusted according to some of the participants' suggestions, by implementing a shaking for the fear emotion.

3 INTERACTION IN VR (LAB STUDY)

Thirdly, we let 42 students (21 female; age: 18 - 55; $M = 24.1$, $SD = 5.59$) interact randomly with one of the companions in a VR game. Subsequently, they filled out the Discrete Emotions Questionnaire (DEQ) [7]. Therefore, we adapted the instructions to ask participants to assess the companion's feelings instead of their own emotions. We formulated five items for the perceived impact of the companion on the participants' emotions (table 2). Independent t-tests reveal that, the positive companion had significantly lower scores in the negative subscales of the DEQ whereas the negative companion had significantly lower scores in the positive subscales (see table 1). Regarding the the companion's effect on the users' emotions, the results show significant differences regarding the positively formulated items, whereby the positive companion reached higher values. Likewise, the group who interacted with the negative companion showed higher values in the negatively formulated items.

4 FUTURE RESEARCH & CONCLUSION

We present the design of two emotionally expressive virtual companions and evaluated the recognizability of its facial expressions as well as it's impact on the participants' emotions in a VR application. Our overarching goal is to find out which emotional displays of virtual characters are more effective in reducing stress. According to the *Emotional Similarity* or *Emotional Contagion* theories both, a distinct or a similar feeling other can be helpful in a stressful situation. Therefore, as next step we will investigate the effect of our two different emotionally expressive companions on the users' emotions, for example during the anticipation of stress.

Table 2: Mean values for the participants' emotional state items. Significant p -values marked with a * ($p < .05$).

Items	PC		NC	
	M		p	
1. C. had an anxiety-reducing effect on me.	4.24	2.86	.002*	
2. C. calmed me down.	3.90	3.00	.047*	
3. C. helped me regulate my emotions.	3.62	2.52	.009*	
4. C.'s behavior increased my anxiety.	1.48	2.48	.011*	
5. C. made me nervous.	2.52	2.67	.745	

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